NE 523 HW4

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Question 1 Laplacian Adjoint

The adjoint of $\mathbf{L}$ is defined using

$$(g, \mathbf{L}f) = \left(f, \mathbf{L}^{\dagger}g\right). \tag{1}$$

Expanding the inner product, the LHS becomes

$$\left(g, \frac{\partial^2 f}{\partial x^2}\right) = \int_{-a}^{a} g(x)f''(x)dx. \tag{2}$$

Applying integration by parts,

$$\left(g, \frac{\partial^2 f}{\partial x^2}\right) = \left[g(x)f'(x)\Big|_{-a}^{a} - \int_{-a}^{a} g'(x)f'(x)dx\right], \tag{3}$$

and expanding once more,

$$\left(g, \frac{\partial^2 f}{\partial x^2}\right) = \left[g(x)f'(x)\Big|_{-a}^{a} - \cancel{g'(x)f(x)\Big|_{-a}^{a}} + \int_{-a}^{a} f(x)g''(x)dx\right], \tag{4}$$

where the second term becomes zero since $f(-a) = f(a) = 0$. Further using the fact that $f'(-a) = -f'(a)$, this becomes

$$\left(g, \frac{\partial^2 f}{\partial x^2}\right) = \left[f'(a)\left(g(a) + g(-a)\right) + \int_{-a}^{a} f(x)g''(x)dx\right] \tag{5}$$

Then, *if* $g(-a) = g(a) = 0$,

$$\left(g, \frac{\partial^2 f}{\partial x^2}\right) = \left(f, \frac{\partial^2 g}{\partial x^2}\right) \tag{6}$$

and thus the operator $\mathbf{L}$ is self-adjoint.

Question 2 Orthogonality of Laplacian Eigenfunctions

2(a) The solution takes the form

$$f(x) = b\cos(\lambda x) + d\sin(\lambda x), \tag{7}$$

and since the solution must be symmtric, $d = 0$ and thus

$$f(x) = b\cos(\lambda x). \tag{8}$$

For the boundary conditions to be satisfied, λa must be an odd-integer multiple of $\pi/2$ and thus the positive eigenvalues are

$$\lambda_n = \frac{(2n+1)\pi}{2a}. \tag{9}$$

where $n = 0, 1, \ldots$. The associated eigenfunctions are

$$f_n(x) = \cos\left(\frac{(2n+1)\pi}{2a}x\right) \tag{10}$$

To simplify notation, let m and ℓ be positive odd integers. Then the inner product of two eigenfunctions is

$$\int_{-a}^{a} \cos\left(\frac{\pi m x}{2a}\right)\cos\left(\frac{\pi \ell x}{2a}\right) dx \tag{11}$$

which can be expressed

$$\frac{1}{2}\int_{-a}^{a} \cos\left(\frac{\pi(m+\ell)x}{2a}\right) + \cos\left(\frac{\pi(m-\ell)x}{2a}\right) dx \tag{12}$$

which is equal to

$$\frac{1}{2}\left[\frac{2}{\pi(m+\ell)}\sin\left(\frac{\pi(m+\ell)x}{2a}\right) + \frac{2}{\pi(m-\ell)}\cos\left(\frac{\pi(m-\ell)x}{2a}\right)\right]_{-a}^{a}. \tag{13}$$

Since $m+\ell$ and $m-\ell$ are both even integers, the terms inside the sine become integer multiples of π at the bounds, and thus the inner product is zero if $m \neq \ell$.

2(b) Since φ can be represented using a linear combination of the eigenfunctions,

$$-\frac{d^2\varphi}{dx^2} = -\sum_{n=0}^{\infty} \alpha_n \frac{d^2 f_n}{dx^2} = \sum_{n=0}^{\infty} \alpha_n \lambda_n^2 f_n(x) = S \tag{14}$$

then the coefficients are found using orthogonality by multiplying by a particular eigenfunction and integrating over the domain,

$$\int_{-a}^{a} \sum_{n=0}^{\infty} \alpha_n \lambda_n^2 f_n(x) f_m(x) dx = \alpha_m \lambda_m^2 \int_{-a}^{a} f_m(x)^2 dx = S \int_{-a}^{a} f_m(x) dx \tag{15}$$

and thus

$$\alpha_m = \frac{S}{\lambda_m^2} \frac{\int_{-a}^{a} f_m(x) dx}{\int_{-a}^{a} f_m(x)^2 dx} \tag{16}$$

where the integral of f finally evaluates to

$$\int_{-a}^{a} f_m(x) dx = \int_{-a}^{a} \cos\left(\frac{(2+1)\pi}{2a} x\right) dx = (-1)^m \frac{4a}{(2m-1)\pi} \tag{17}$$

and

$$\int_{-a}^{a} f_m(x)^2 dx = a, \tag{18}$$

so

$$\alpha_m = (-1)^m \frac{S}{\lambda_m^2} \frac{4}{(2m+1)\pi} \tag{19}$$

and the solution is

$$\varphi(x) = \sum_{n=1}^{\infty} \alpha_n f_n(x). \tag{20}$$

2(c) The analytical solution is

$$\varphi(x) = \frac{S}{2}\left(a^2 - x^2\right). \tag{21}$$

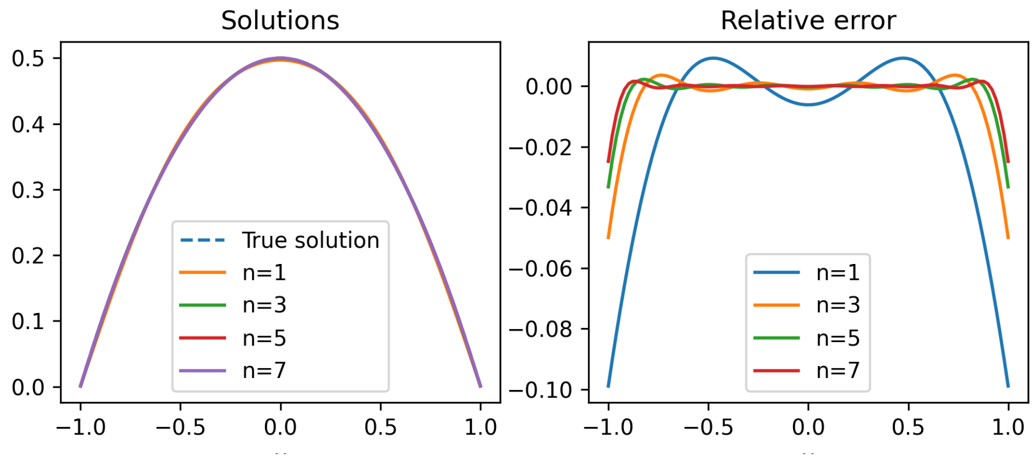

Figure 1: Solutions and Errors

The truncated consine reconstructions are shown in Figure 1, where $S = 1$ and $a = 1$.